

Question: 1

Convert the following IP address to its 32-bit binary representation: 172.16.5.10.
Write out each octet in binary, separated by dots?

Answer: 1. Decimal to Binary Conversion per Octet

172 $\rightarrow$ 10101100

16 $\rightarrow$ 00010000

5 $\rightarrow$ 00000101

10 $\rightarrow$ 00001010

2. Complete 32-Bit Binary Representation

10101100.00010000.00000101.00001010

Question: 2

Given the binary IP address 11000000.10101000.00000001.00001111, convert it back to standard dotted decimal notation?

Answer: 1. Binary to Decimal Conversion per Octet

11000000 $\rightarrow$ 128 + 64 = 192

10101000 $\rightarrow$ 128 + 32 + 8 = 168

00000001 $\rightarrow$ 1 = 1

00001111 $\rightarrow$ 8 + 4 + 2 + 1 = 15

2. Complete Dotted-Decimal Notation

192.168.1.15

Question: 3

You are given a list of 10 IP addresses used by different apps (e.g., 192.168.43.1, 8.8.8.8, 10.0.5.25, 172.20.14.2, 203.0.113.5, 172.15.255.255, 192.0.2.10, 192.168.1.100, 123.45.67.89, 172.16.0.1). For each, identify if it is a private or public IP address and explain your reasoning?

Answer: 1. 192.168.43.1 \(\rightarrow\) Private

Reasoning: It falls within the Class C private IP range (192.168.0.0 - 192.168.255.255), commonly used for local mobile hotspots.

2. 8.8.8.8 \(\rightarrow\) Public

Reasoning: It sits outside all reserved private ranges. It is a globally unique public IP address owned by Google for its public DNS service.

3. 10.0.5.25 \(\rightarrow\) Private

Reasoning: It falls within the Class A private IP range (10.0.0.0 - 10.255.255.255) used for large internal enterprise networks.

4. 172.20.14.2 \(\rightarrow\) Private

Reasoning: It falls within the Class B private IP range (172.16.0.0 - 172.31.255.255).

5. 203.0.113.5 \(\rightarrow\) Public (Documentation)

Reasoning: It falls outside the private IP blocks. It belongs to a globally reserved public block (TEST-NET-3) used specifically for documentation and examples.

6. 172.15.255.255 \(\rightarrow\) Public

Reasoning: Even though it starts with 172, it falls outside the private Class B range because 15 is lower than the starting private limit of 16.

7. 192.0.2.10 \(\rightarrow\) Public (Documentation)

Reasoning: It sits outside the 192.168.x.x private range. It belongs to a public block (TEST-NET-1) reserved for textbook examples and documentation.

8. 192.168.1.100 $\rightarrow$ Private

Reasoning: It falls within the Class C private IP range (192.168.0.0 - 192.168.255.255), typical for standard home routers.

9. 123.45.67.89 $\rightarrow$ Public

Reasoning: It is completely outside any private IP scope and is globally routable on the public internet.

10. 172.16.0.1 $\rightarrow$ Private

Reasoning: It is the exact starting address boundary of the Class B private IP range (172.16.0.0 - 172.31.255.255).

Question: 4

For the CIDR notations /20, /25, and /30, calculate the total number of usable IP addresses in each subnet. Show your working for each calculation.
Hint: Usable addresses = ($2^{\text{number of host bits}}$) minus 2 (for network and broadcast addresses)?

Answer: 1. Calculation for /20 Subnet

Host Bits: 32 total bits (-20) network bits $(=12)$ host bits.

Total IP Addresses: $(2^{12} = 4,096)$.

Usable IP Addresses: $(4,096 - 2 =)$ 4,094 usable addresses.

2. Calculation for /25 Subnet

Host Bits: 32 total bits (-25) network bits $(=7)$ host bits.

Total IP Addresses: $(2^7 = 128)$.

Usable IP Addresses: $(128 - 2 =)$ 126 usable addresses.

3. Calculation for /30 Subnet

Host Bits: 32 total bits $\setminus(-30\setminus)$ network bits $\setminus(=2\setminus)$ host bits.

Total IP Addresses: $\setminus(2^2 = 4\setminus)$.

Usable IP Addresses: $\setminus(4 - 2 = \setminus)$ 2 usable addresses.

Question: 5

Imagine you are setting up a small LAN for a gaming night with friends using your WiFi router at home. Assign static private IP addresses to 5 devices (laptop, phone, tablet, smart TV, and PlayStation) using the 192.168.0.0/24 range. List the IP you would assign to each device and explain why you chose that range?

Answer: 1. Static IP Allocations (192.168.0.0/24 Range)

Default Gateway (WiFi Router): 192.168.0.1 (Reserved for the router to manage the local network traffic).

Gaming Laptop: 192.168.0.10

PlayStation: 192.168.0.11

Smart TV: 192.168.0.20

Smart Phone: 192.168.0.30

Tablet: 192.168.0.31

2. Why This Specific Range Was Chosen

Private Network Design: The 192.168.0.0/24 block belongs to the Class C private IP address range. These addresses are strictly reserved for local area networks (LANs) and do not route to the public internet, preventing external hackers from directly connecting to your gaming devices.

Subnet Capacity: The /24 subnet mask (255.255.255.0) provides 254 usable host addresses (from .1 to .254). This gives plenty of room for your 5 friends' devices without any risk of IP address exhaustion.

Ease of Management: Using low, sequential static blocks (like .10 and .11 for primary gaming rigs) keeps the network organized. It ensures the router's automatic DHCP server won't accidentally give these addresses to other guest devices, preventing IP address conflicts during your gaming session.